

# ABSTRACT

---

In recent years, the integration of Internet of Things (IoT) technology into food quality monitoring systems has emerged as a pivotal innovation in ensuring the freshness and safety of produce. This project focuses on leveraging IoT architectures to monitor the quality of vegetables and fruits using an embedded dual-sensor array. Ethylene gas, a primary byproduct of organic matter decomposition, serves as a direct chemical indicator of spoilage, while ambient temperature and humidity heavily influence physical deterioration rates.

The system operates through an automated routine: the targeted produce is placed inside a containment environment where an MQ-4 gas sensor continuously captures analog chemical markers alongside digital climate data tracking from a DHT11 environmental sensor. An Arduino Uno edge microcontroller processes these concurrent data streams against precise thresholds to instantly evaluate freshness states. If parameters are normal, the system displays a "Fresh" status. If thresholds are breached, the produce is classified as "Spoiled," instantly activating a local physical buzzer and displaying the real-time alert on the second row of a 16x2 LCD layout.

Simultaneously, the processed metrics and safety logs are transmitted over secure HTTP GET requests to the cloud-based ThingSpeak IoT platform via an ESP8266 Wi-Fi module. This unified framework enables continuous remote monitoring, analytical graphing, and proactive alerts across the supply chain. Ultimately, this approach provides stakeholders with immediate, data-driven insights to optimize inventory management, maximize consumer food safety, extend product shelf life, and systematically minimize post-harvest agricultural waste.

---

**Keywords:** Internet of Things (IoT), Arduino Uno, MQ-4 Gas Sensor, DHT11, ESP8266 Wi-Fi Module, ThingSpeak Cloud, Food Quality Monitoring, Spoilage Alert System, Buzzer Interfacing.